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Polarisation mode dispersion correlations with the coarse-step method

Christos Braimiotis *, Marc Eberhard, Keith Blow

Photonics Research Group, Aston University, Birmingham B4 7ET, UK

Received 10 March 2005; received in revised form 31 October 2005; accepted 23 December 2005

Abstract

Having a fixed differential-group delay (DGD) term b' in the coarse-step method results in a repetitive pattern in the autocorrelation function (ACF). We solve this problem by inserting a varying DGD term at each integration step. Furthermore we compute the range of values needed for b' and simulate the phenomenon of polarisation mode dispersion for different statistical distributions of b' . We examine systematically the modified coarse-step method compared to the analytical model, through our simulation results.

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Keywords: Coarse-step method; Polarisation mode dispersion; Differential-group delay

1. Introduction

The phenomenon of polarisation mode dispersion in optical fibres is normally modelled using the coarse-step method. This approach is based on the idea of replacing the continuous random birefringence by a sequence of randomly connected fibres each of which has a constant birefringence. It has recently been demonstrated [1] that this method produces an autocorrelation function (ACF), which is qualitatively different to the one calculated from a continuous analytical model. It is important to stress that the phenomenon has been shown to exist with respect to emulators, that is, experimentally concatenating pieces of polarisation maintaining fibres with polarisation scattering at the beginning of each section but not in the context of numerical simulations, with the widely used coarse-step method.

The coarse-step method uses a constant differential-group delay (DGD) term which is applied after each integration step. This results in the polarisation ACF containing a series of harmonics instead of a single peak.

In this paper we show that by allowing the DGD to vary from step-to-step the harmonic artifacts can be removed.

In particular, we investigate the reduction of the side peaks as we gradually increase the number of different values from which we choose the DGD per length. We also allow the pool of random values to vary between a uniform and a Gaussian distribution and numerically we confirm that there is no difference in the long length regime. In the next section we will refer to the coarse-step method. We will then discuss the calculation and properties of the ACF. Finally we show numerical simulations which demonstrate the production of the correct ACF.

2. Review of the coarse-step method

The coarse-step method describes the phenomenon of the polarisation mode dispersion (PMD) as a two-step process. As a first step we have the splitting of the field in the two orthogonal modes due to the difference in their propagation constants. The second step allows the power coupling between the two modes, which is physically caused by fibre imperfections. For a more thorough understanding of the coarse-step method one needs to examine the CNLS equation as explained in [2]

$$i \frac{\partial \Psi}{\partial z} + i b' \sigma_3 \frac{\partial \Psi}{\partial t} - \frac{1}{2} \beta'' \frac{\partial^2 \Psi}{\partial t^2} + n_2 k_0 \left[\frac{5}{6} |\Psi|^2 \Psi + \frac{1}{6} (\Psi^\dagger \sigma_3 \Psi) \sigma_3 \Psi \right] = 0, \quad (1)$$

* Corresponding author. Tel.: +44 0121 204 3693.

E-mail address: braimioc@aston.ac.uk (C. Braimiotis).

where β'' is the group-velocity dispersion parameter and n_2 is the nonlinear Kerr coefficient. The group delay per unit length is denoted as $b' = (\beta'_1 - \beta'_2)/2$. The prime indicates differentiation with respect to the angular frequency and β_1, β_2 are the two propagation constants while σ_3 is the third Pauli matrix. The purpose of this study was purely to simulate the PMD phenomenon with the coarse-step method. Accordingly we set $\beta'' = 0$ and $n_2 = 0$ so that there is no chromatic dispersion or Kerr non-linearity and we include in our simulations only the second term which causes the linear PMD. In order to introduce the power coupling between the modes we multiply at each step the field with a rotational matrix as in Eq. (3)

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (2)$$

$$S = \begin{pmatrix} \cos \alpha & \sin \alpha \exp(i\phi) \\ -\sin \alpha \exp(i\phi) & \cos \alpha \end{pmatrix}, \quad (3)$$

where α is the random angle used for the mixing of the two modes and ϕ indicates the phase. Within the context of the coarse-step method we numerically introduce the splitting of the field as the two modes propagate with different velocities in the fibre by multiplying with the DGD term in the frequency domain. Simulations then with the coarse-step method give additional peaks in the ACF.

3. Autocorrelation function of the coarse-step method

The analytical ACF is given by the following equation [3,4], where $\Delta\tau$ is the mean DGD

$$\text{ACF}_{\text{analytical}} = \frac{3}{\Delta\omega^2} \left[1 - \exp\left(\frac{-\Delta\tau^2 \Delta\omega^2}{3}\right) \right] \quad (4)$$

In our implementation we calculate the ACF by averaging over the different initial conditions as in [1]

$$\text{ACF}_{\text{numerical}} = \frac{|\langle \Omega(\omega) \cdot \Omega(\omega_0) \rangle|}{|\langle \Omega(\omega_0) \cdot \Omega(\omega_0) \rangle|} \quad (5)$$

$\Omega(\omega)$ is the polarisation dispersion vector.

The periodicity of the ACF of the coarse-step method results numerically, from the splitting of the field to the two orthogonal polarisation modes which is the effect of the DGD per unit length.

It is convenient at this stage to represent the initial pulse as a delta function. Successively, after a number of simulation steps, and neglecting the power coupling which does not affect our case, the time domain picture will consist of a series of delta functions all of which are separated by a constant multiple of time determined by the constant parameter b' . Thus the autocorrelation of the series of delta functions in the frequency domain, will result in another series of regularly spaced delta functions.

Furthermore we can monitor the reduction of the side-peaks by having b' given as a variable chosen from a pool of random values that exist within a certain width. By gradually increasing the width of the distribution it is possible

to measure the reduction of the first harmonic against the centre peak.

4. Numerical simulations

The simulations were performed for a system having the following characteristics $D_{\text{PMD}} = 3 \text{ ps}/(\text{km})^{1/2}$, correlation length $L_c = 100 \text{ m}$, the integration step was 1 km while the optical bandwidth of the simulation was 4 THz represented by a resolution of 4096 sampling points. As can be seen in Fig. 1, using a fixed value DGD we obtain the periodic pattern in the ACF of the coarse-step method which deviates from the proposed analytic model as given in Eq. (4).

Continuously we use the “refined” coarse-step method. The value of b' is chosen from a range of random numbers spaced equally around the mean value of $3 \text{ ps}/\text{km}$. As presented in Fig. 2 the harmonics decrease as we draw more

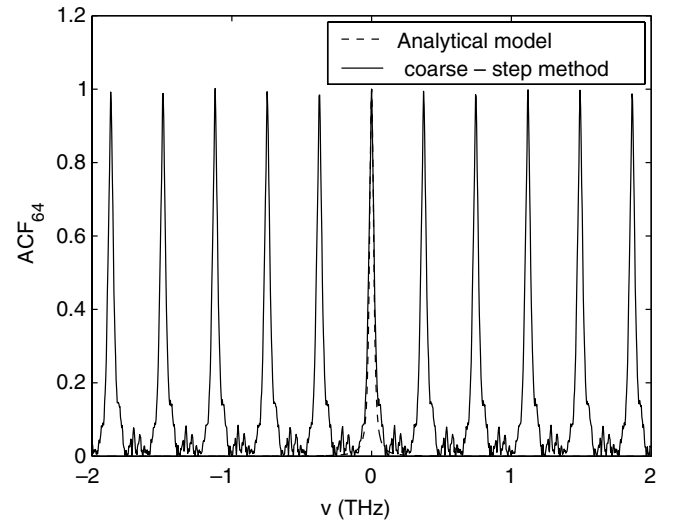


Fig. 1. ACF of the coarse-step method with a fixed DGD of $3 \text{ ps}/\text{km}$ for 64 km compared to the analytical model of Eq. (4).

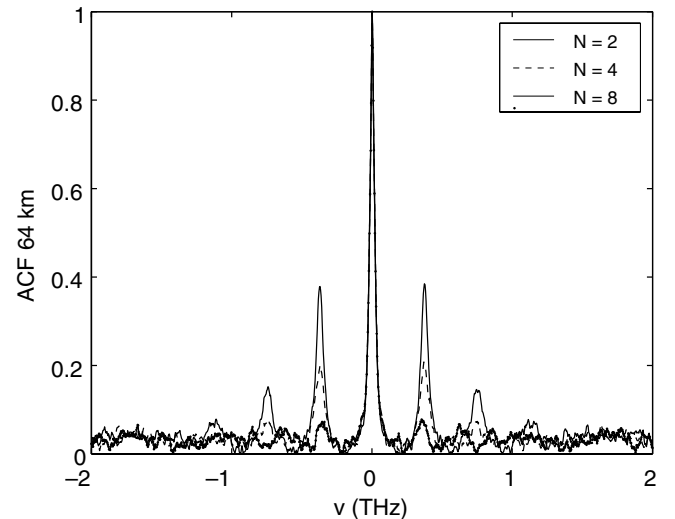


Fig. 2. ACF simulations of the coarse-step method for an increasing number of values for the DGD per length, $N = 2, 4, 8$.

numbers for the value of b' and thus increase the range of choice. Initially we used a number $N=2$ for b' , choosing between 2.8 and 3.2 ps. Progressively number N is increased to 4 numbers 2.6, 2.8, 3.2 and 3.4 ps and finally we reach $N=8$. The DGD in the last case varies between the values 2.2, 2.4, 2.6, 2.8, 3.2, 3.4, 3.6 and 3.8 ps, thus having a range of 1.6 ps around the average value of 3 ps/km.

As shown in Fig. 3 it is obvious that the autocorrelation background will reduce to 10% when we reach a value of 1 ps for the width $\Delta b'$ of the random pool. We now use the modified coarse-step method for the option of using a Gaussian distribution for the random pool of $|b'|$. It is important to note that since the absolute value is used, the negative numbers for the birefringence that in any case

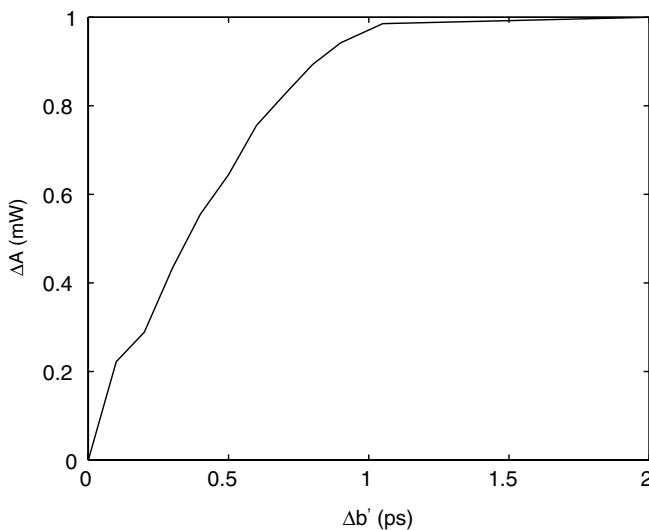


Fig. 3. Evolution of the difference in amplitude between the fundamental frequency and the first harmonic for an increasing width of the random pool spaced evenly around an average b' of 3 ps/km.

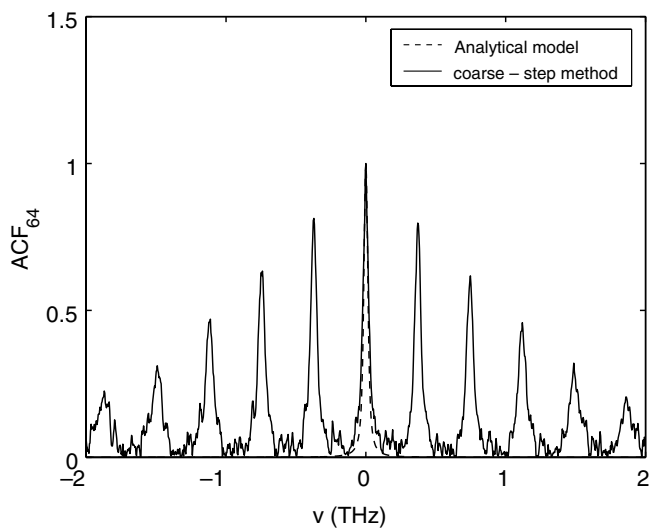


Fig. 4. ACF of the coarse-step method with DGD varying according to a Gaussian distribution of $\sigma = 0.05$ ps and mean $\mu = 3$ ps compared to the analytical model of Eq. (4), the transmission length was 64 km.

have no physical meaning are discarded. In Fig. 4 we show that, even for a small variance for $|b'|$ of 0.05 ps, the harmonics of the ACF become suppressed. An important point to mention is that the reduction on the harmonics of the ACF is obtained in [5]. The authors there indicate that extra randomisation introduced by setting the DGD per length, of an emulator, from a Gaussian distribution leads to removing the periodicity gradually by increasing σ , the standard deviation of the distribution. The variance of $|b'|$ is increased to 1 ps in Fig. 5 where there are no longer any harmonics present and the central peak still agrees very well with the analytical prediction of Eq. (4). As the simulations have been performed in the long-length regime, different random distributions will produce the same effect.

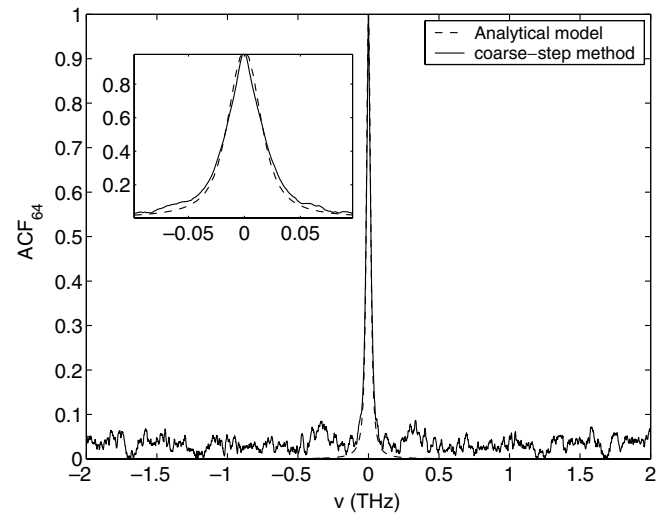


Fig. 5. ACF of the coarse-step method with DGD varying according to a Gaussian distribution of $\sigma = 1$ ps and mean $\mu = 3$ ps compared to the analytical model given by Eq. (4), the transmission length was 64 km.

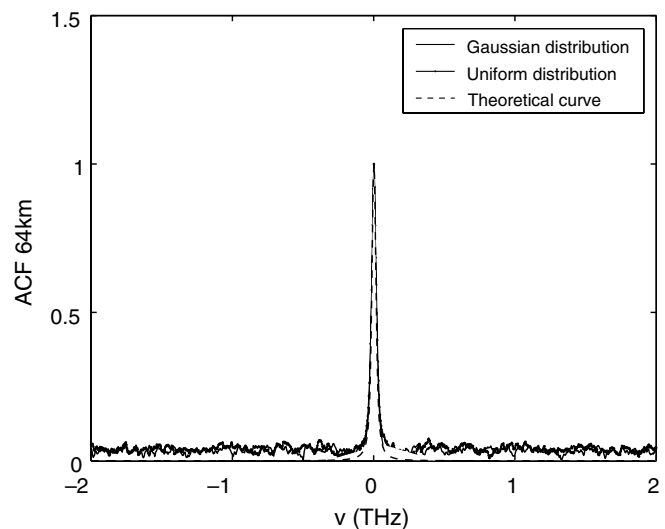


Fig. 6. Results for the ACF of two different type of distributions, a uniform and a Gaussian.

This is demonstrated in Fig. 6 where a uniform and a Gaussian distribution of the same range produce the same effect on the harmonic artifacts. It is worth noting that the PDF of the DGD does not change significantly when we modify the coarse-step method, as it is shown in Fig. 7 thus it is correct to use the coarse-step method with a varying differential group delay model rather than the conventional one. Further we demonstrate results that are produced from introducing only two different values for b' . From Figs. 8–11 where we used randomly, one of two values ($N=2$) for b' at each step, it is clear that the effect on the resulting ACF is multiplicative. Using only two values for b' though, results in harmonics existing within the

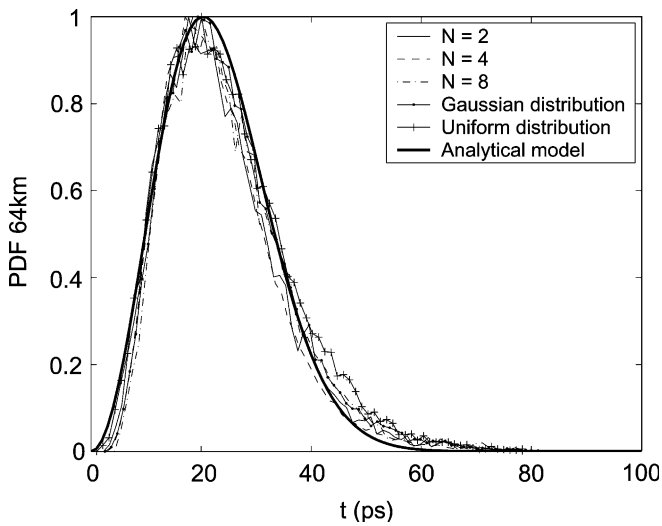


Fig. 7. Comparison of the PDF simulations of the refined coarse-step method with a varying DGD. Results are presented for $N=2, 4, 8$ a uniform and a Gaussian distribution.

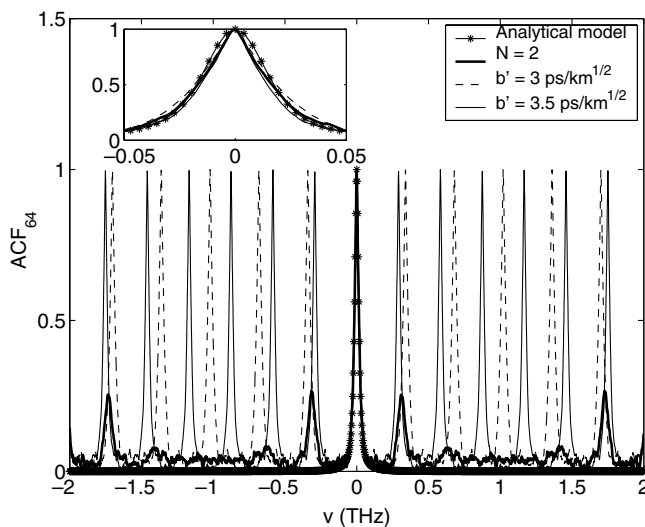


Fig. 8. Results for the ACF, choosing $b' = 3$ ps or $b' = 3.5$ ps randomly at each step and comparing with the individual ACF for each of these two values.

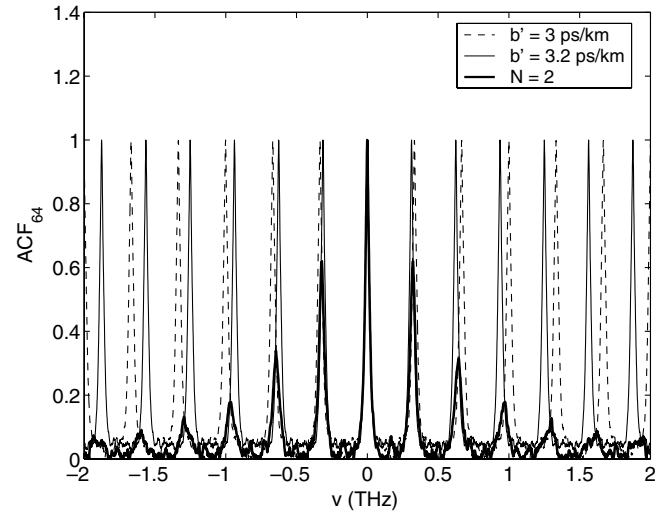


Fig. 9. Results for the ACF, choosing $b' = 3$ ps or $b' = 3.2$ ps randomly at each step and comparing with the individual ACF for each of the two values.

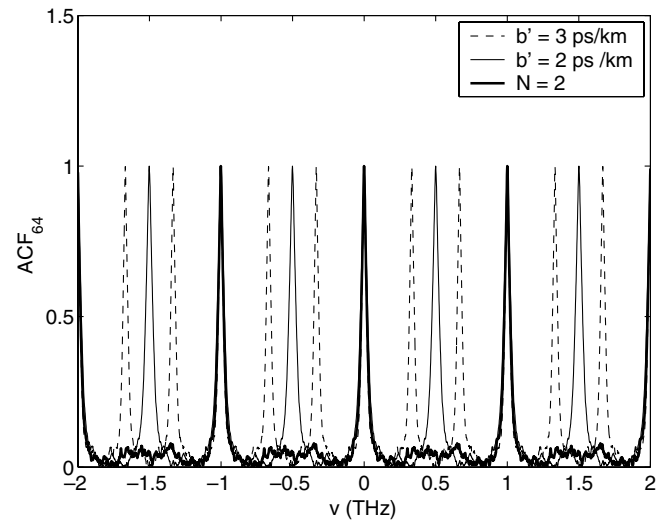


Fig. 10. Results for the ACF, choosing $b' = 3$ ps or $b' = 2$ ps randomly at each step and comparing with the individual ACF for each of the two values.

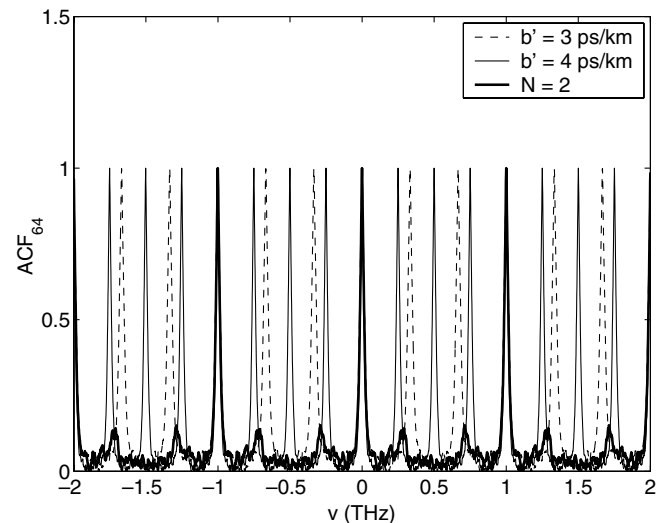


Fig. 11. Results for the ACF, choosing $b' = 3$ ps or $b' = 4$ ps randomly at each step and comparing with the individual ACF for each of the two values.

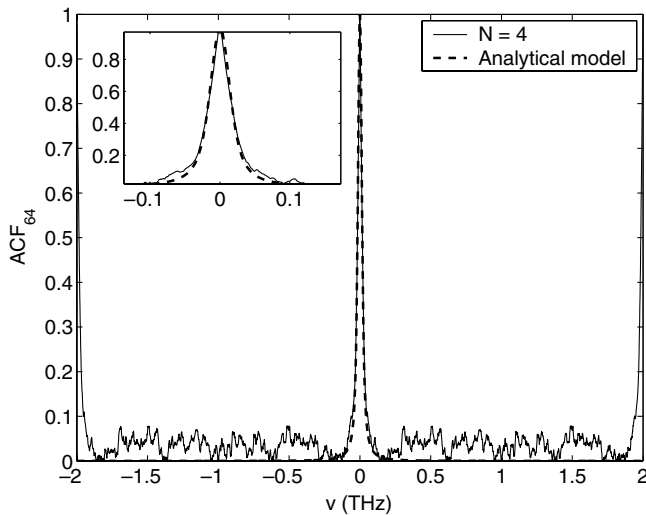


Fig. 12. Results for the ACF, choosing b' from the values 2, 4, 2.5, 3.5 ps randomly at each step and comparing with the analytical curve.

bandwidth of interest. A better choice is to use 4 values ($N = 4$) as in Fig. 12 where the harmonics are totally eliminated within the frequency range of 4 THz. Thus instead of using a constant b' . Or randomising the DGD per unit length by recalculating this value at every step, based on the strategy of inserting a statistical distribution and gradually increasing the random pool of numbers we choose from, as in [5]. It was proven more numerically efficient to precalculate four numbers, that meet at higher frequencies, in the beginning of the simulation, and choose between these values at random, at every step to enlarge

the bandwidth of transmission in order to produce a simulator that can accommodate WDM systems.

5. Conclusion

Using the coarse-step method for simulating PMD, results in additional side-peaks in the ACF. Moreover unwanted autocorrelation artifacts have been known to exist within the context of emulators. The importance of such an effect has not been taken into account, to our knowledge, for the case of numerical simulations since by definition the coarse-step method is used with a fixed b' . Refining the coarse-step method by using a range of values for b' chosen from a distribution or from a pool of numbers is as physically adequate as using a constant DGD per unit length but it satisfies the conditions as given in Lima et al. [6] namely the PDF being a Maxwellian and the ACF tending quadratically toward zero.

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